Graphing linear inequalities in two variables

1

a No **b** Yes

2

a

i No **ii** No **iii** No **iv** Yes

b

i No **ii** No **iii** No **iv** Yes

c

i No **ii** No **iii** No **iv** Yes

3

a $y \leq 5$ **b** $x \geq 2$ **c** $y \geq -2x - 6$ **d** $y < 2x + 6$

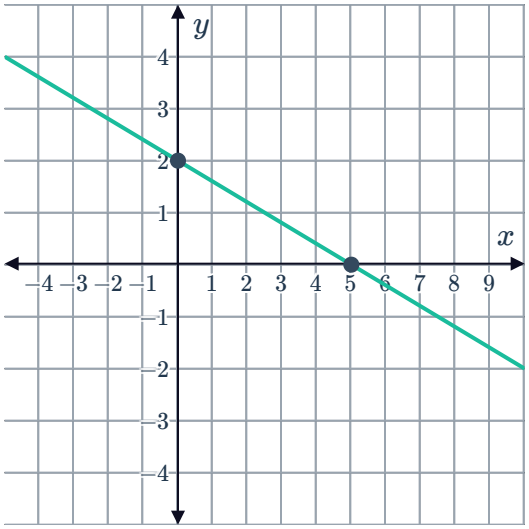
e $y > -3x$ **f** $y \leq 2x + 5$ **g** $y \geq 3x - 3$ **h** $y < -2x - 4$

i $y \leq -2$ **j** $y > -3x + 6$

4

a $x = 5$ **b** $y = 2$

c



d $2x + 5y > 10$

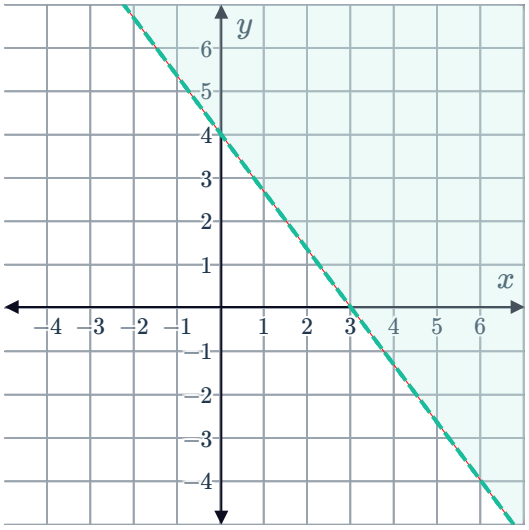
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5

a $x = 3$

b $y = 4$

c



d No

6

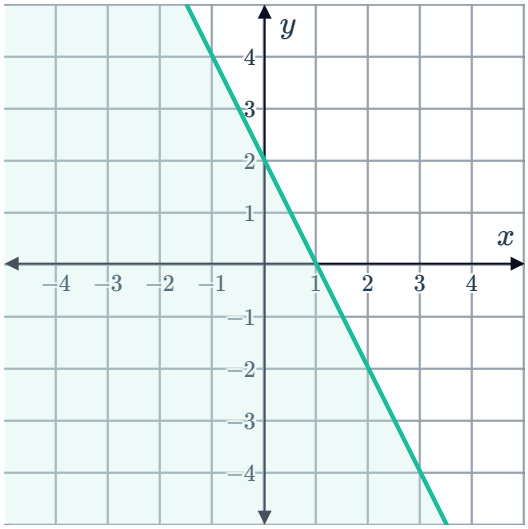
a $x = 1$

b $y = 2$

c

i No **ii** Yes **iii** No **iv** No

d



e Yes

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7

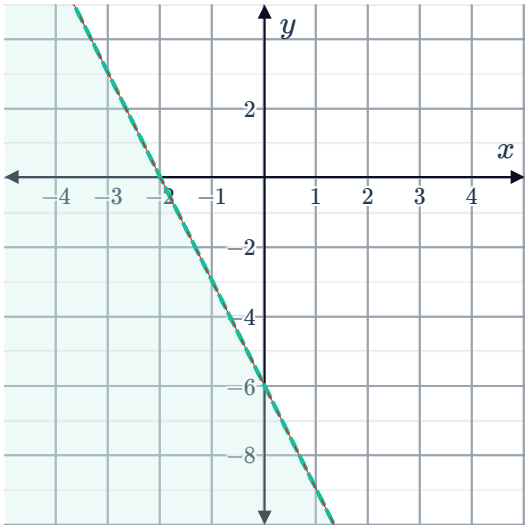
a $x = -2$

b $y = -6$

c

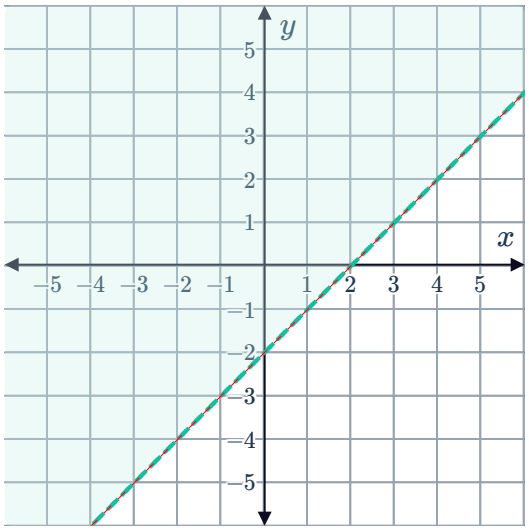
i No **ii** No **iii** Yes **iv** No

d

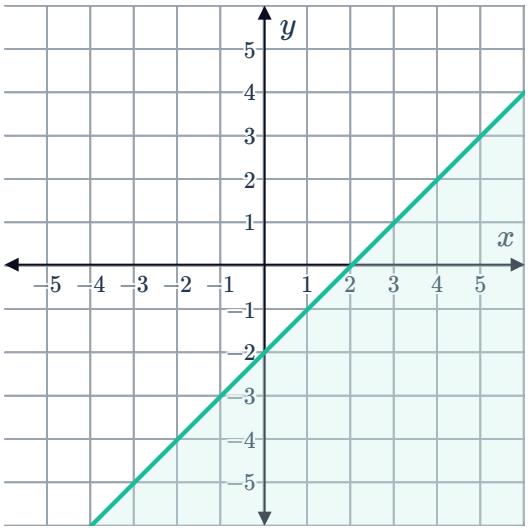


8

a



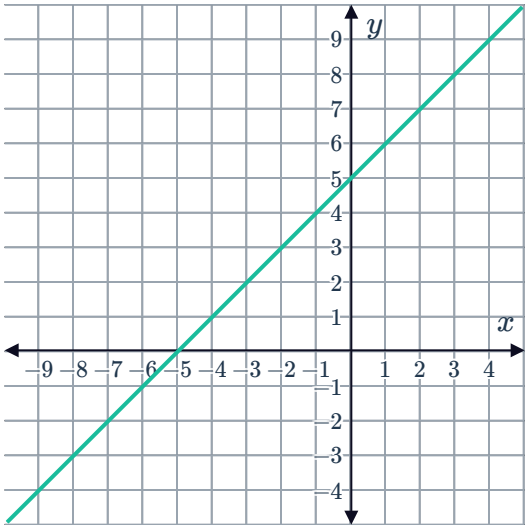
b



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9

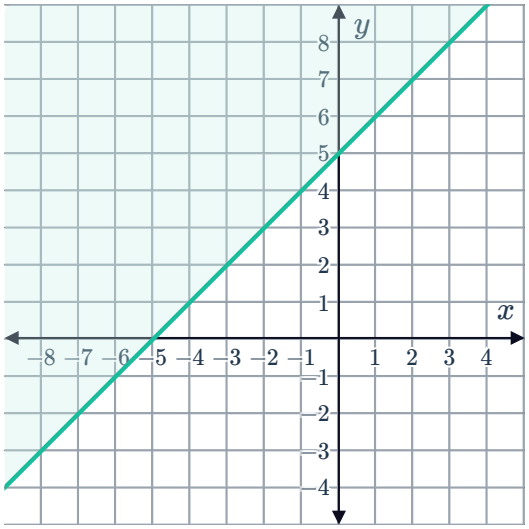
a



b

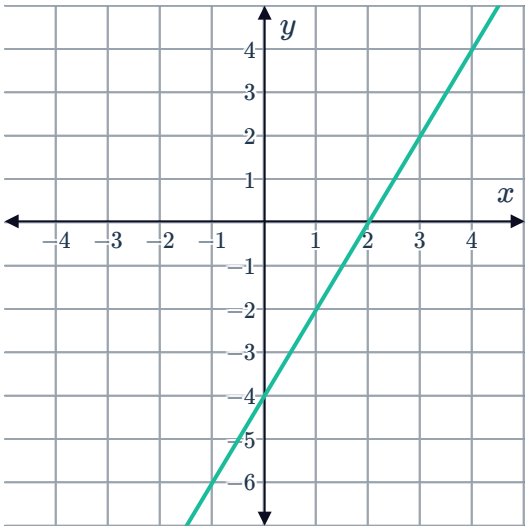
i Yes **ii** No

c



10

a

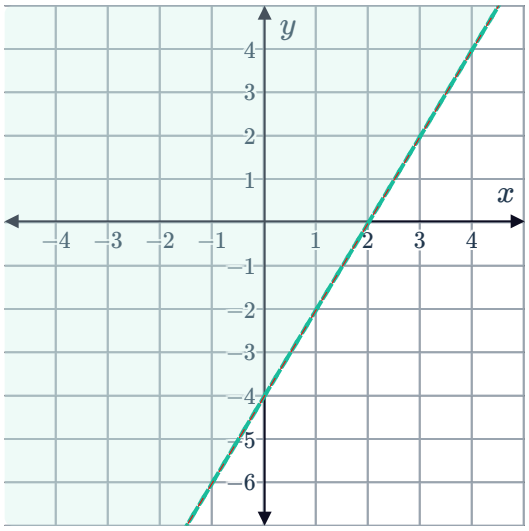


b

i No **ii** Yes

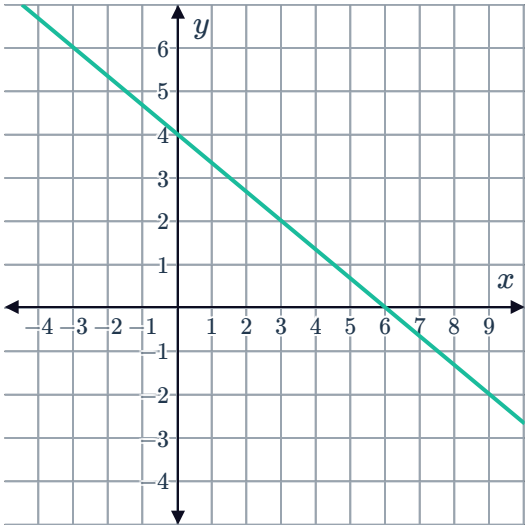
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c



11

a

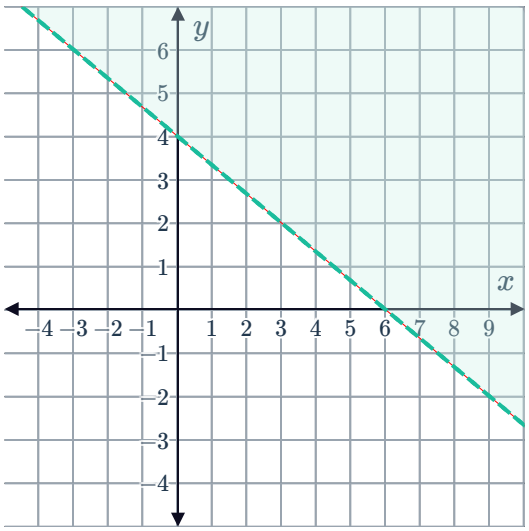


b

i No

ii Yes

c



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12

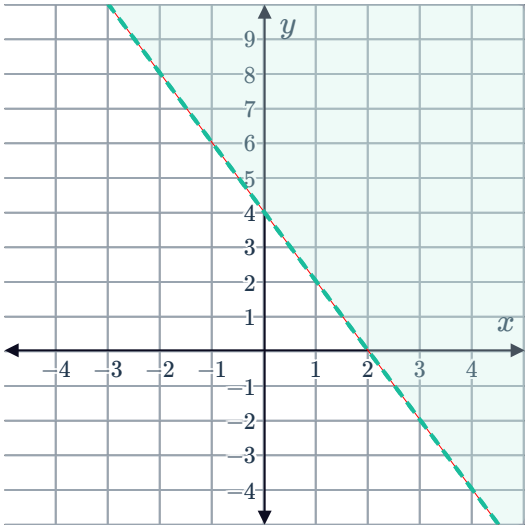
a

x	0	1	2
y	4	2	0

b

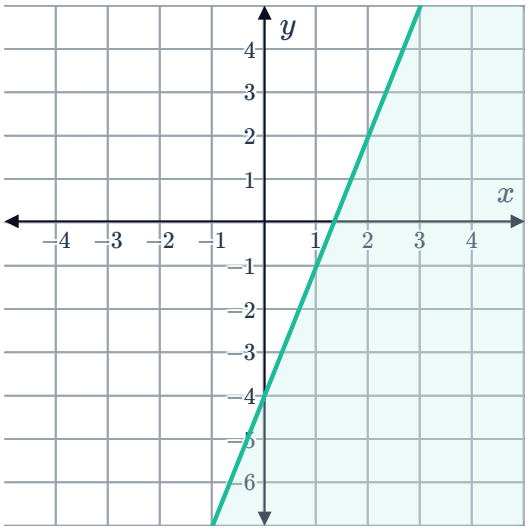
i No ii Yes iii No iv No

c

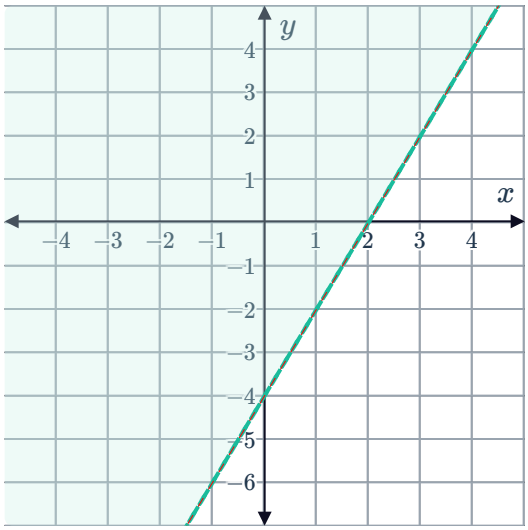


13

a

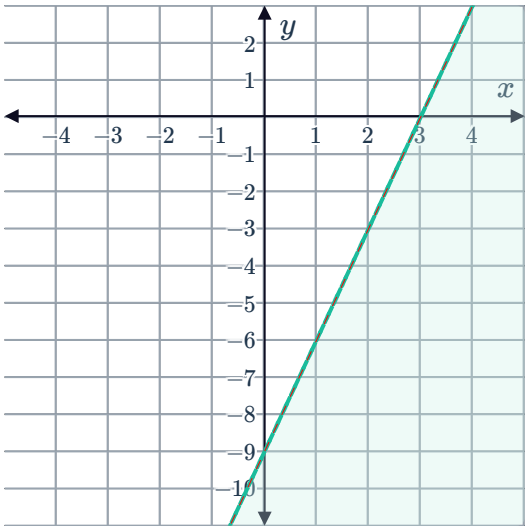


b

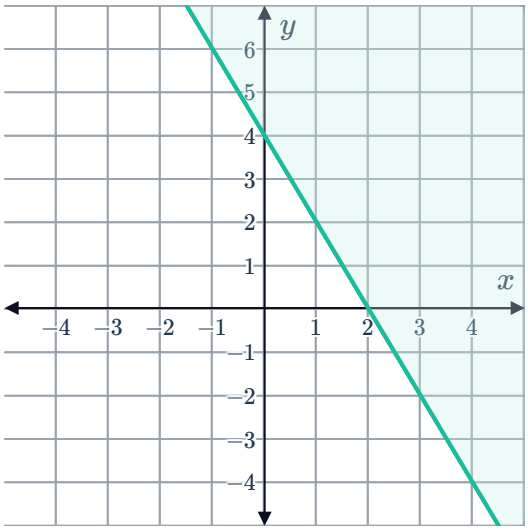


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c



d



14

a

i 4 ii $y = 4x - 5$ iii $y \leq 4x - 5$

b

i 3 ii $y = 3x - 6$ iii $y \geq 3x - 6$

15 $y \leq 95 - \frac{19}{9}x$

16 $y \leq 75 - \frac{15}{17}x$

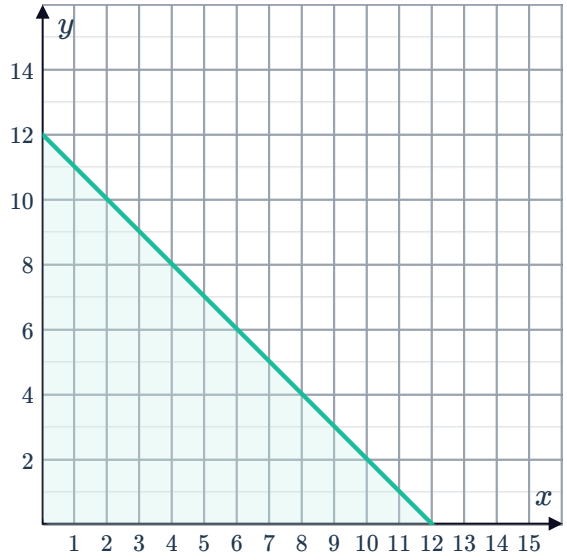
17 $y \leq x - 58$

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18

a $y \leq 12 - x$

b



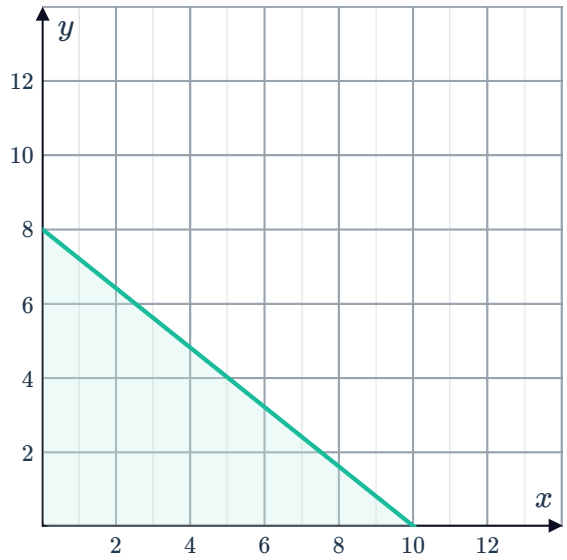
c

i No **ii** Yes **iii** No **iv** No

19

a $y \leq 8 - \frac{4}{5}x$

b



c

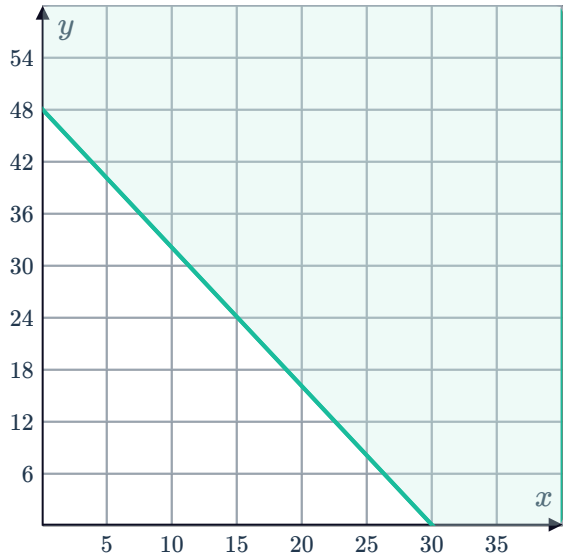
i No **ii** Yes **iii** No **iv** No

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20

a $y \geq 48 - \frac{8}{5}x$

b



c

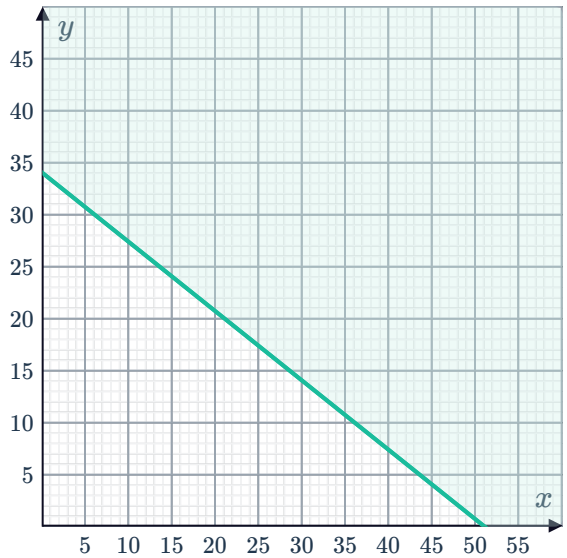
i No **ii** No **iii** No **iv** Yes

d 30 books

21

a $y > 34 - \frac{2}{3}x$

b



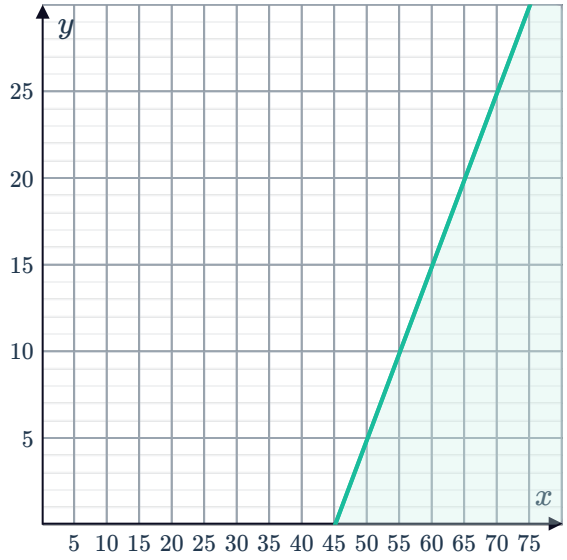
c **B**

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22

a $y \leq x - 45$

b



c 65 minutes

Additional questions

23

a Solid **b** Dashed **c** Solid **d** Dashed

24

a $(2, 4)$ **b** $(2, 4)$, $(-3, 2)$ and $(-1, -6)$

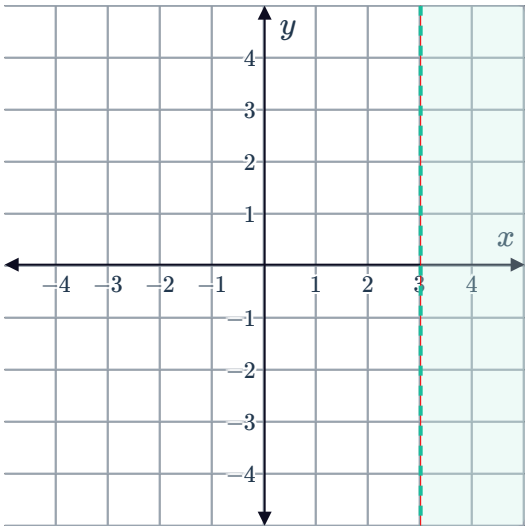
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25

a

i The point $(0, 0)$ does not satisfy the inequality.

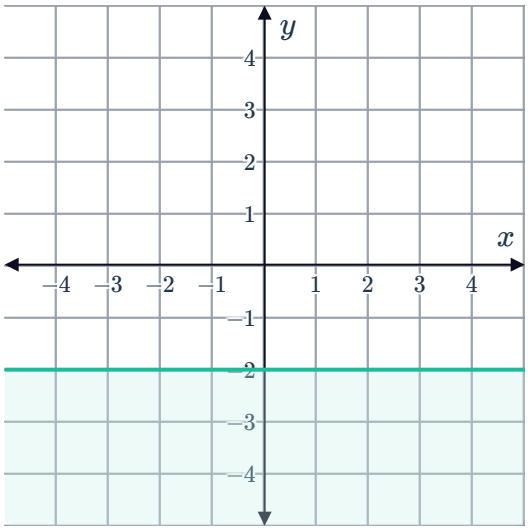
ii



b

i The point $(0, 0)$ does not satisfy the inequality.

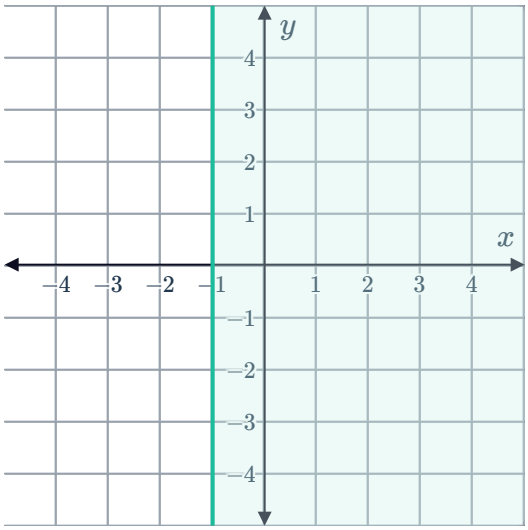
ii



c

i The point $(, 0, 0)$ satisfies the inequality.

ii



26 $x + y \geq 43$

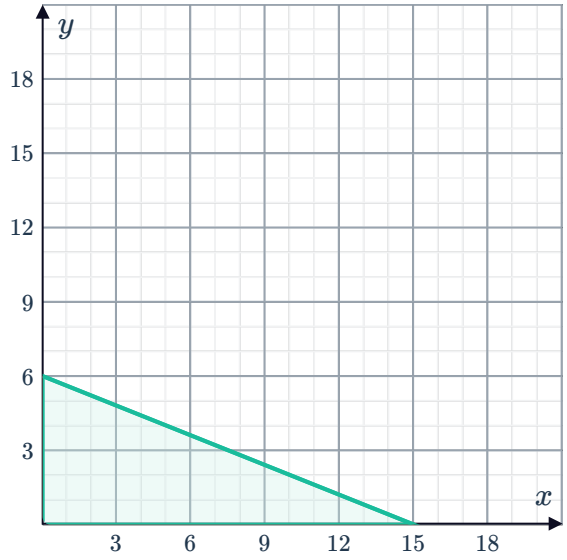
27 $10x + 13y \geq 260$

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28

a $2x + 5y \leq 30$

b



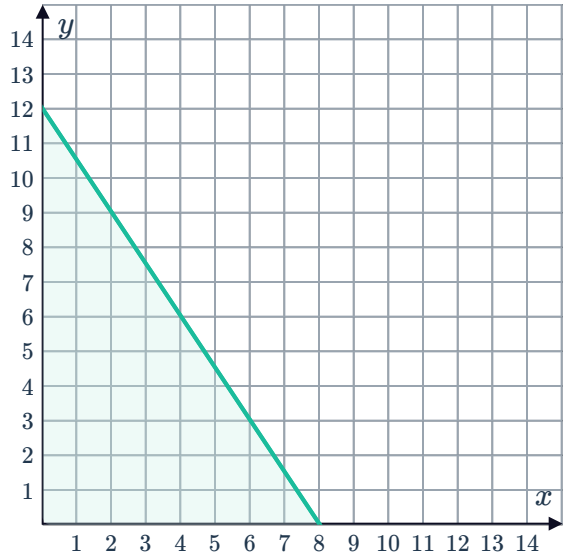
c The x-intercept of 15 represents the greatest number of games of Table Tennis that Oprah could play if she decides not to play any games of Frog Days.

d The y-intercept of 6 represents the greatest number of games of Frog Days that Oprah could play if she decides not to play any games of Table Tennis.

29

a $6x + 4y \leq 48$

b



c There are 61 different ways, one for each integer coordinate inside the solution set of the inequality.

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